

📘 AI/ML Internship – Day 72 Notes & Tasks

🟢 Module 9: Similarity Search – Understanding How RAG Finds the Right Information

📌 Part 1: Recap of Day 71

Yesterday we learned:

- ✅ Build a Basic PDF Question Answering Chatbot
- ✅ Load PDFs
- ✅ Chunk Documents
- ✅ Generate Embeddings
- ✅ Store Embeddings in ChromaDB
- ✅ Ask Questions from PDFs

Today we will learn:

- What is Similarity Search?
 - Why Similarity Search is Needed
 - How Similarity Search Works
 - Distance Metrics
 - Cosine Similarity
 - Euclidean Distance
 - Dot Product
 - Top-K Retrieval
-

📌 Part 2: Why Do We Need Similarity Search?

Suppose your PDF has 500 pages.

A user asks:

What is the internship duration?

Should the AI read all 500 pages?

✗ No.

Instead, it searches for the most relevant chunks and only sends those chunks to the LLM.

This process is called Similarity Search.

📌 Part 3: What is Similarity Search?

Definition

Similarity Search is the process of comparing the user's question with all stored document embeddings and retrieving the most relevant information based on meaning, not just matching keywords.

Simple Definition

Instead of searching for the exact words, the AI searches for similar meanings.

Example

Document:

The AI Internship lasts for 3 months.

User asks:

How long is the internship?

The words are different, but the meaning is the same.

Similarity Search retrieves the correct chunk.

📌 Part 4: Keyword Search vs Similarity Search

Keyword Search	Similarity Search
Matches exact words	Matches meaning
Misses synonyms	Understands similar sentences

Keyword Search

Similarity Search

Traditional search engines Modern AI applications

Less accurate

More accurate

Example

Question:

Where is my package?

Document:

Your order has been shipped.

Keyword Search:

✗ May fail because the words differ.

Similarity Search:

✓ Retrieves the correct document because it understands the meaning.

✦ Part 5: How Similarity Search Works

User Question

|



Convert Question into Embedding

|



Compare with Stored Embeddings

|



Calculate Similarity Scores

|



Retrieve Top Matching Chunks



Send Context to LLM



Generate Final Answer

Part 6: Distance Metrics

Similarity between vectors is calculated using mathematical techniques called distance metrics.

The three most common metrics are:

- Cosine Similarity
 - Euclidean Distance
 - Dot Product
-

Part 7: Cosine Similarity

Definition

Cosine Similarity measures the angle between two vectors.

If two vectors point in the same direction, they have similar meanings.

Range

- 1 → Exactly similar
 - 0 → Not related
 - -1 → Opposite meaning
-

Example

Sentence A:

I love Artificial Intelligence.

Sentence B:

I enjoy studying AI.

Cosine Similarity:

0.95

Since the value is close to 1, the meanings are very similar.

Part 8: Euclidean Distance

Definition

Euclidean Distance measures the straight-line distance between two vectors.

Interpretation

- Small Distance → More Similar
- Large Distance → Less Similar

Example:

If two vectors are very close together, they likely represent sentences with similar meanings.

Part 9: Dot Product

The Dot Product compares vectors by multiplying their corresponding values.

Some embedding models use the dot product because it is efficient and works well with normalized vectors.

Part 10: Top-K Retrieval

Instead of retrieving only one result, RAG systems usually retrieve the Top-K most relevant chunks.

Example:

User Question:

Explain the internship duration.

Retrieved Chunks:

1. Internship duration is 3 months.
2. Internship starts on August 1.
3. Certificate is provided after completion.

The LLM combines all relevant chunks to generate a better answer.

📌 Part 11: Similarity Search using ChromaDB

```
results = collection.query(  
    query_texts=["What is the internship fee?"],  
    n_results=3  
)
```

```
print(results)
```

Explanation

- `query_texts` → User's question
 - `n_results=3` → Retrieve the top 3 similar document chunks
-

📌 Part 12: Similarity Search using FAISS

```
D, I = index.search(query_embedding, k=3)
```

Explanation

- `D` → Distance or similarity scores
 - `I` → Index positions of retrieved vectors
 - `k=3` → Retrieve the top 3 matches
-

✦ Part 13: Real-World Applications

AI Chatbots

Find answers from company documents.

Healthcare

Retrieve treatment guidelines.

Banking

Search policy documents.

E-Commerce

Recommend similar products.

Education

Answer questions from textbooks and notes.

✦ Part 14: Advantages

- ✓ Understands meaning instead of exact words
 - ✓ Handles synonyms and paraphrased questions
 - ✓ Improves chatbot accuracy
 - ✓ Essential for modern RAG systems
 - ✓ Works well with large document collections
-

✦ Part 15: Limitations

- ✗ Depends on high-quality embeddings
- ✗ Poor chunking can reduce retrieval accuracy

✖ Very large datasets require optimized indexing

🔴 Part 16: Theory Questions

1. What is Similarity Search?

Similarity Search compares embeddings to retrieve the most relevant information based on semantic meaning.

2. Why is Similarity Search important in RAG?

It retrieves the most relevant document chunks, which provide context for the LLM to generate accurate responses.

3. What is Cosine Similarity?

Cosine Similarity measures the angle between two vectors to determine how similar their meanings are.

4. What is Euclidean Distance?

Euclidean Distance measures the straight-line distance between vectors. Smaller distances indicate greater similarity.

5. What is Top-K Retrieval?

Top-K Retrieval returns multiple relevant document chunks instead of only one.

🔴 Part 17: Interview Questions

Explain Similarity Search.

Similarity Search retrieves information by comparing the semantic meanings of embeddings rather than matching exact keywords.

Why is Similarity Search better than Keyword Search?

Because it understands the meaning of sentences, allowing it to retrieve relevant information even when different words are used.

Name three similarity metrics.

- Cosine Similarity
 - Euclidean Distance
 - Dot Product
-

Why do RAG systems retrieve multiple chunks?

To provide enough context for the LLM to generate complete and accurate answers.

What does `n_results=3` mean in ChromaDB?

It retrieves the top three most similar document chunks for the user's query.

Tasks for Day 72

Theory

1. Define Similarity Search.
 2. Differentiate Keyword Search and Similarity Search.
 3. Explain Cosine Similarity.
 4. What is Top-K Retrieval?
 5. Why is Similarity Search essential in RAG?
-

Practical Task 1

Create a ChromaDB collection and store at least 10 sample text documents.

Practical Task 2

Perform similarity searches for the following questions:

- What is the internship fee?
- How long is the internship?
- Is placement assistance available?

Observe the retrieved results.

Practical Task 3

Change the value of `n_results` to 1, 3, and 5, and compare the outputs.

Practical Task 4

Create a comparison table for:

- Cosine Similarity
- Euclidean Distance
- Dot Product

Include their advantages and common use cases.

Practical Task 5

Write a Python program that accepts a user query and displays the Top 3 most similar document chunks from ChromaDB.